StrataWave Teknik Dogrulama Raporu

DEEPSOIL-benzeri is akisina ve OpenSees destekli effective-stress dogrulamasina sahip hibrit analiz platformu icin mevcut repo kanitlarinin derlenmis ozeti.

Amac ve Kapsam

Bu rapor, StrataWave reposunda bugun mevcut olan kanitlari toplar ve urunun teknik durumunu dis paylasima uygun sekilde belgelendirir.
Rapor yeni bir urun ozelligi eklemez; mevcut example, benchmark, parity ve dokumantasyon artifactlerini kullanir.

Mimari Ozet

Native solver yollari: linear, eql, nonlinear.
Effective-stress kaniti: OpenSees adapter yolu ve ilgili local run artifactleri.
Arayuz katmanlari: React/FastAPI web UI, Streamlit engineering UI, CLI, Python SDK.
Veri depolama ve raporlama: results.h5, results.sqlite, campaign/parity JSON ve Markdown artifactleri.

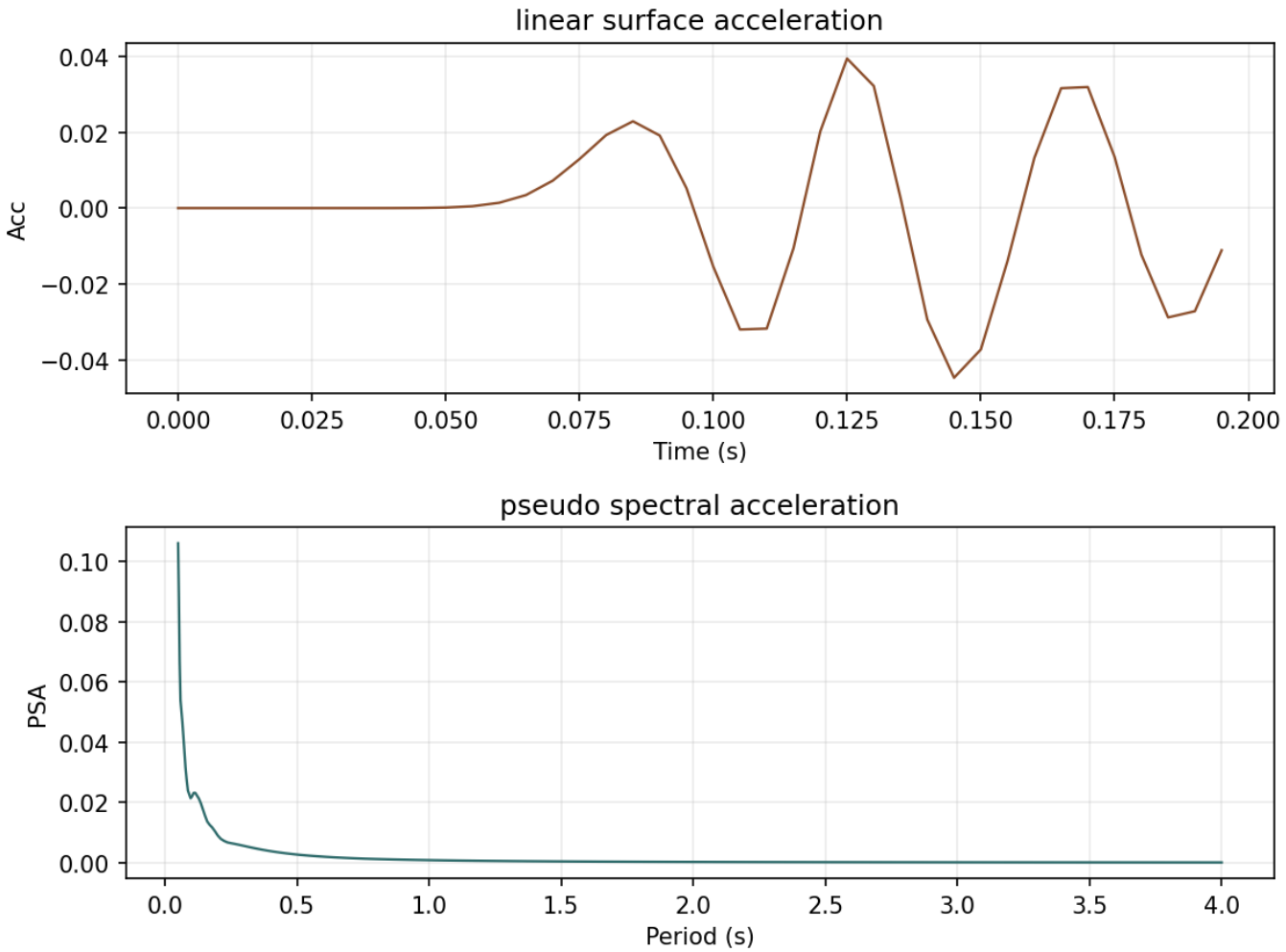
Sonuc Hukmu

Calisiyor: native lineer, EQL, nonlinear ornekler ve OpenSees effective-stress adapter kosusu local evidence ile mevcuttur.
Kismi: DEEPSOIL parity, UI parity ve scientific confidence derinligi.
Eksik: native effective-stress solver ve tam publication-grade parity kapanisi.

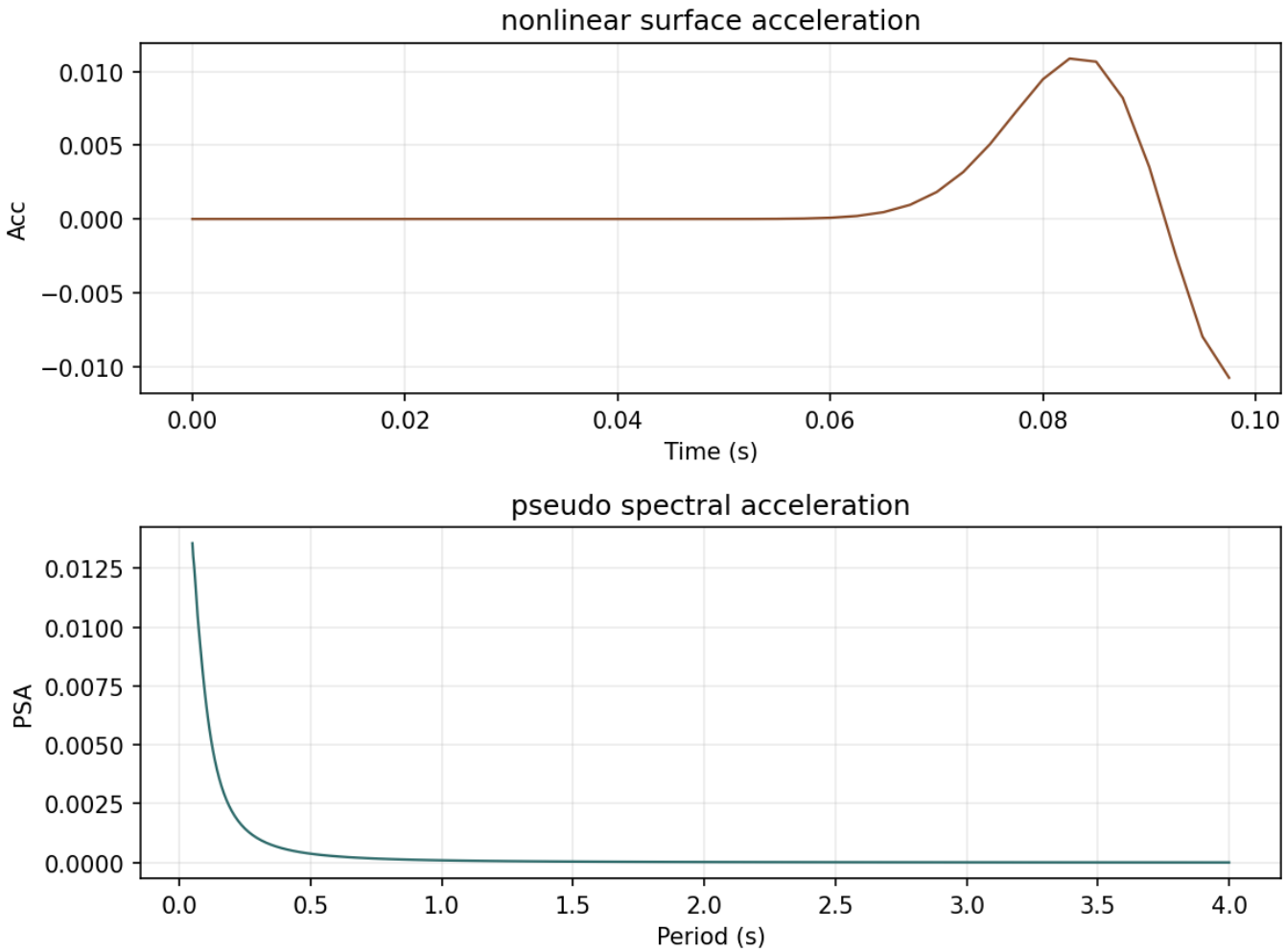
Durum Matrisi

Native linear solver path: done | Smoke pack and local campaign outputs are used as evidence.
Native equivalent-linear solver path: done | EQL behavior is evidenced by the reference pack and local campaign outputs.
Native nonlinear MKZ/GQH path: done | This is native total-stress nonlinear behavior, not native effective-stress FE.
OpenSees effective-stress adapter: done | The adapter path is proven by local run artifacts; native effective-stress solver is still absent.
Darendeli calibration workflow: done | Calibration inputs and a local run exist; published-reference calibration closure is still pending.
DEEPSOIL parity tooling: done | Local parity benchmark currently reports ran=6, skipped=0; full dedicated-runner parity is still pending.
React/FastAPI DEEPSOIL-like UI: partial | Workflow parity exists, but full DEEPSOIL UI parity is not claimed.
Release signoff and scientific confidence: done | Scientific confidence exists as a repo artifact; it is not equivalent to full external publication validation.

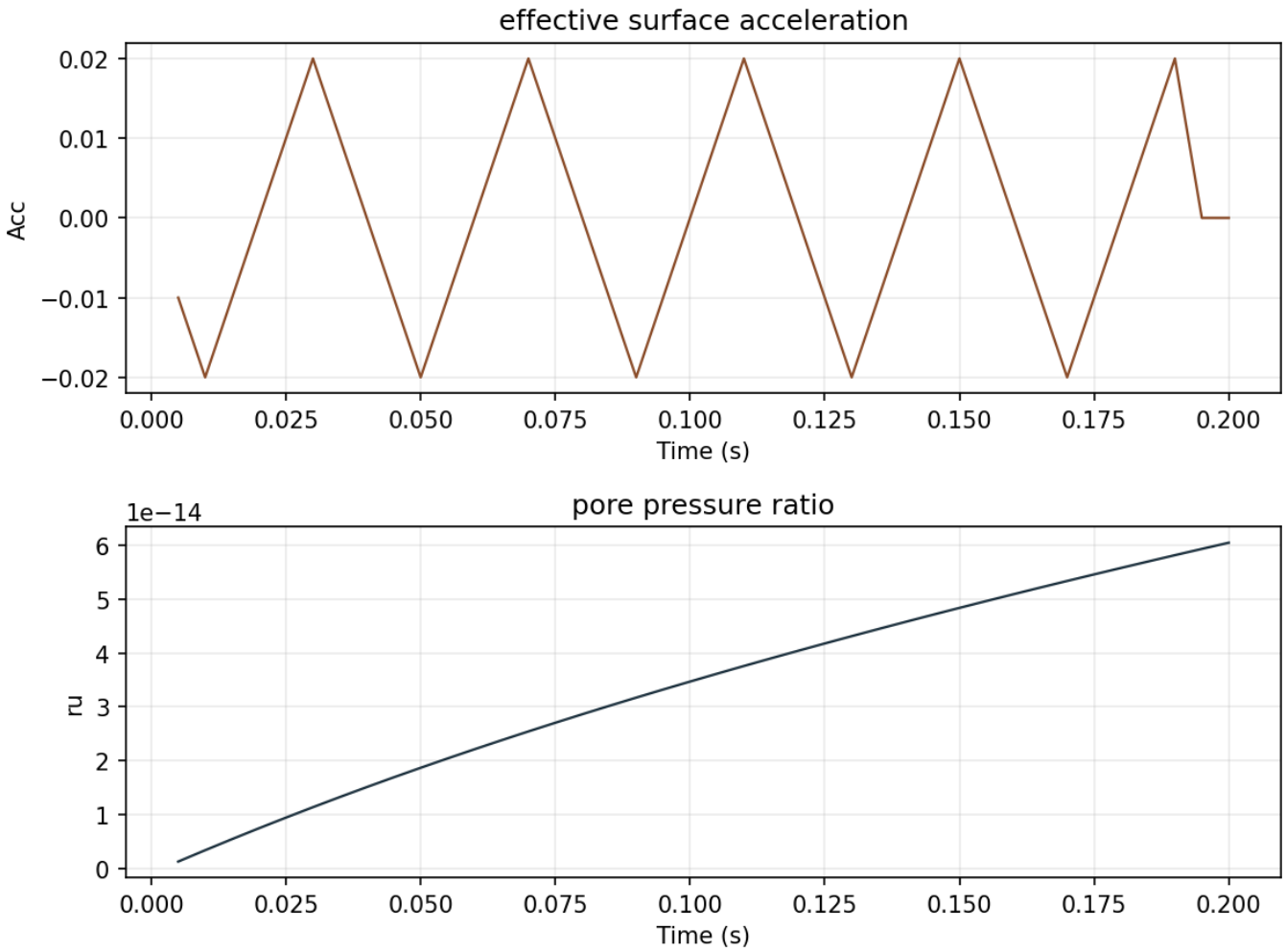
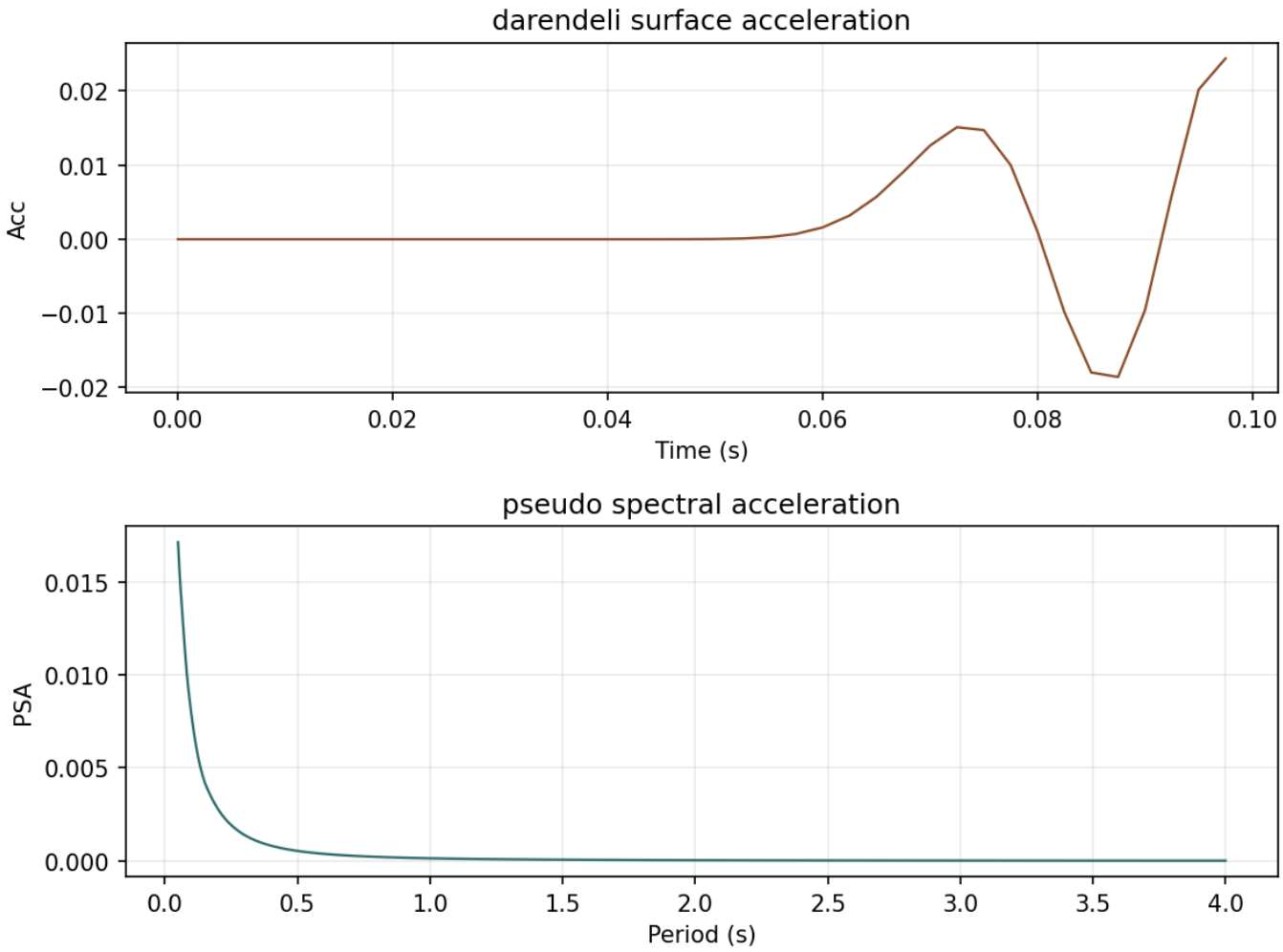
Ek Grafik - linear_overview



Ek Grafik - nonlinear_overview



Ek Grafik - effective_stress_overview



Ek Grafik - darendeli_overview